

Frequency analysis via copulas: Theoretical aspects and applications to hydrological events

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[1] In this paper we provide a general theoretical framework exploiting copulas for studying the return periods of hydrological events; in particular, we consider events depending upon the joint behavior of two nonindependent random variables, an approach which can easily be generalized to the multivariate case. We show that using copulas may greatly simplify the calculations and may even yield analytical expressions for the isolines of the return periods, both in the unconditional and in the conditional case. In addition, we show how a new probability distribution may be associated with the return period of specific events and introduce the definitions of sub-, super-, and critical events as well as those of primary and secondary return periods. An illustration of the techniques proposed is provided by analyzing some case studies already examined in literature. **INDEX TERMS:** 3210 Mathematical Geophysics: Modeling; 1869 Hydrology: Stochastic processes; 1854 Hydrology: Precipitation (3354); 1821 Hydrology: Floods; **KEYWORDS:** 2-Copulas, return period

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1. Introduction

[2] The return period of a given event is usually defined as the average time elapsing between two successive realizations of the event itself. In the applications, and especially in hydrology, the design of works usually adopt the return period of a prescribed event as a common criterion for sizing the manufactures: indeed, the return period provides a very simple, yet efficient, means for doing risk analysis because it is able to concentrate into a single number a large amount of information.

[3] Historically, *Hiemstra et al.* [1976] did pioneering work in frequency analysis; a thorough review can be found by *Singh* [1987]. In most situations the analysis of the return period involves univariate cases; unfortunately, this may lead to an over/underestimation of the risk associated with a given event (see, e.g., *Raynal-Villasenor and Salas* [1987] and *De Michele et al.* [2004] for a practical example; see also section 3.1 for a thorough discussion). As a matter of fact, hydrological events are often characterized by the joint behavior of several random variables (RV), and these are usually nonindependent: to mention a few examples, consider the storm duration and intensity [*De Michele and Salvadori*, 2003; *Salvadori and De Michele*, 2004], the flood peak and volume [*De Michele et al.*, 2004], drought events characterized by duration-magnitude-intensity [see *Kim et al.*, 2003, and references therein], and so on. Other significant works concerning the application of joint probability distributions to model extreme hydrological events

are by *Yue et al.* [1999], *Yue* [2000], and *Yue et al.* [2001]. As a consequence, the relevant events should better be defined in terms of two or more variables; this makes things complicated since the family of interesting events increases with the number of variables.

[4] In the following we shall mainly concentrate on the bivariate case. In bivariate frequency analysis it is not obvious what should be understood by an event with a given return period (nor it is in multivariate analysis); unfortunately, the literature on the subject is relatively sparse (for a thorough review, see *Yue and Rasmussen* [2002, and references therein]), and in some studies, several concepts were even misused [see *Yue*, 2001]. Clearly, incorrect interpretation of these concepts will lead to misinterpretation of frequency analysis results; for these reasons, we shall try and give a clear illustration of them below.

[5] The problem of specifying a probability model for dependent bivariate observations can be greatly simplified by expressing the corresponding joint distribution F_{XY} in terms of its marginals, F_X and F_Y , and an associated dependence function, C , implicitly defined through the functional identity $F_{XY} = C(F_X, F_Y)$. As will be shown in section 2, the function C is a suitable 2-Copula (hereinafter referred to as copula); unfortunately, the use of copulas in hydrology as well as in other geophysical and environmental sciences is still marginal. The interesting point is that the properties of F_{XY} can be discussed in terms of the structure of C . In fact, it is precisely the copula that “captures” the features of a joint distribution. In addition, measures of association and dependence properties between RVs can be more easily investigated in terms of copulas.

[6] Contrary to traditional approaches (for which extensions to the multivariate case often are not clear), our approach is easy to generalize to an n -dimensional framework. Moreover, quite a few multivariate distributions present in literature are direct extensions of well-known univariate ones and suffer from several limitations and constraints: for instance, the marginal distributions may belong to the same probability family, and the parameters of the marginals may also rule the dependence between the variables considered. On the contrary, our approach does not exhibit such drawbacks. In addition, although the techniques we shall outline later will be illustrated considering only those events best describing the extreme behavior of the phenomena investigated, they can be applied as well to a wider class of hydrologically interesting events. Furthermore, we shall also study events involving the dynamics of one variable conditionally on the values taken by the other.

[7] A further advantage of using copulas is that complex marginal distributions (such as finite mixtures [Titterton et al., 1985], which are receiving increasing attention by practitioners (see section 4.3)) can easily be plugged into the model of interest. Incidentally, we observe that all the bivariate distributions present in literature can be described in a straightforward way in terms of suitable copulas (for an exhaustive list of models see, e.g., Hutchinson and Lai [1990] and Nelsen [1999, and references therein]). Concerning the problems of simulating copulas and fitting copulas to experimental data, see, e.g., many of the references in Nelsen [1999]; other references can be found in the bibliographies of De Michele and Salvadori [2003] and Favre et al. [2004].

[8] In section 3 we shall provide the link between copulas and the return period of bivariate events. As a consequence, instead of considering a particular joint distribution F_{XY} with well-specified marginals F_X and F_Y (as is usual in many applied works, which, in our opinion, denotes a limited view and prospect), we shall calculate the return periods of the events of interest using a suitable copula. This will solve the problem for a wider class of distributions than the function F_{XY} alone (actually, for all the joint laws generated by the copula). We also stress that in some cases it may be even possible to derive the analytical expressions of the isolines of the return periods, both in the unconditional and in the conditional case, and this represents an important (if not essential) piece of information in the applications. In addition, we shall show how a new, well-defined probability distribution may be associated with the return period of specific events; this will lead to the definitions of sub-, super-, and critical events as well as to those of primary and secondary return periods.

[9] Finally, in section 4 we shall illustrate an application of the techniques outlined in the paper by analyzing some case studies already examined in literature, all involving RVs of the highest interest in hydrology. As a result, the investigation of the problems will be simplified, and new information about the phenomena considered will be easily gained.

[10] In order to be clear, in the following we shall present all the statements and the results using a mathematical style involving definitions and results. On the one hand, this will make the presentation concise and well organized; on the

other hand, it will help the reader to identify the relevant formulas as well as the most significative results.

2. An Overview of Copulas

[11] As already mentioned in section 1, a possible way of investigating bivariate data consists of studying the dependence function and the marginals separately. Since copulas exactly describe and model the dependence structure between random variables, independently of the marginal laws involved, we shall limit our analysis just to copulas; however, we shall also discuss the role played by the marginals and will show how to take them into proper account.

[12] In this section we briefly outline the mathematics of copulas needed in the following; all the theoretical justifications can be found by Nelsen [1999] (see also Sempi [2003] for a survey). Let us start off with Definition 1 of copulas, as follows.

[13] • Let $\mathbf{I} = [0, 1]$; a copula is a bivariate function $\mathbf{C}: \mathbf{I} \times \mathbf{I} \rightarrow \mathbf{I}$ such that (1) (uniform marginals) for all $u, v \in \mathbf{I}$ it holds $\mathbf{C}(u, 0) = 0$, $\mathbf{C}(u, 1) = u$, $\mathbf{C}(0, v) = 0$, and $\mathbf{C}(1, v) = v$; (2) (2-increasing) for all $u_1, u_2, v_1, v_2 \in \mathbf{I}$ such that $u_1 \leq u_2$ and $v_1 \leq v_2$ it holds $\mathbf{C}(u_2, v_2) - \mathbf{C}(u_2, v_1) - \mathbf{C}(u_1, v_2) + \mathbf{C}(u_1, v_1) \geq 0$.

[14] The link between copulas and bivariate distributions is provided by the following result [Sklar, 1959]. Hereinafter, F_X, F_Y (F_U, F_V respectively) will denote the marginal distribution functions of the RVs X, Y (U, V , respectively) and Ran will denote their range (Result 1).

[15] • Let F_{XY} be a joint distribution function with marginals F_X and F_Y . Then there exists a copula $\mathbf{C}$ such that

$$F_{XY}(x, y) = \mathbf{C}(F_X(x), F_Y(y)) \quad (1)$$

for all reals x, y . If F_X, F_Y are continuous, then $\mathbf{C}$ is unique; otherwise, $\mathbf{C}$ is uniquely defined on $\text{Ran}(F_X) \times \text{Ran}(F_Y)$. Conversely, if $\mathbf{C}$ is a copula and F_X, F_Y are distribution functions, then F_{XY} given by equation (1) is a joint distribution function with marginals F_X and F_Y .

[16] Although no theoretical constraints exist on the choice of F_X, F_Y we shall limit our investigation to continuous, strictly increasing marginals, this case being the one of greatest interest in the applications; however, only minor changes (involving the use of suitable inverses or pseudo-inverses (see, e.g., Definition 3 below)) are required in case F_X, F_Y do not satisfy such an assumption. Thus by virtue of Result 1, the copula representation will always be unique. This latter assumption has further important consequences. In fact, since copulas are invariant under strictly increasing transformations of X and Y , thanks to the probability integral transform, we may restrict our attention to the pair of RVs (U, V) given by

$$\begin{cases} U = F_X(X) \\ V = F_Y(Y) \end{cases} \quad (2)$$

Clearly, U and V are uniform on $\mathbf{I}$, i.e., $U \sim \mathcal{U}(0, 1)$ and $V \sim \mathcal{U}(0, 1)$, and the pair (U, V) has the same copula as the pair (X, Y) , i.e., $(U, V) \sim \mathbf{C}_{UV} = \mathbf{C}_{XY}$. This is easy to show given the invariance of copulas and the fact that F_X, F_Y in equation (2) are monotone and increasing.

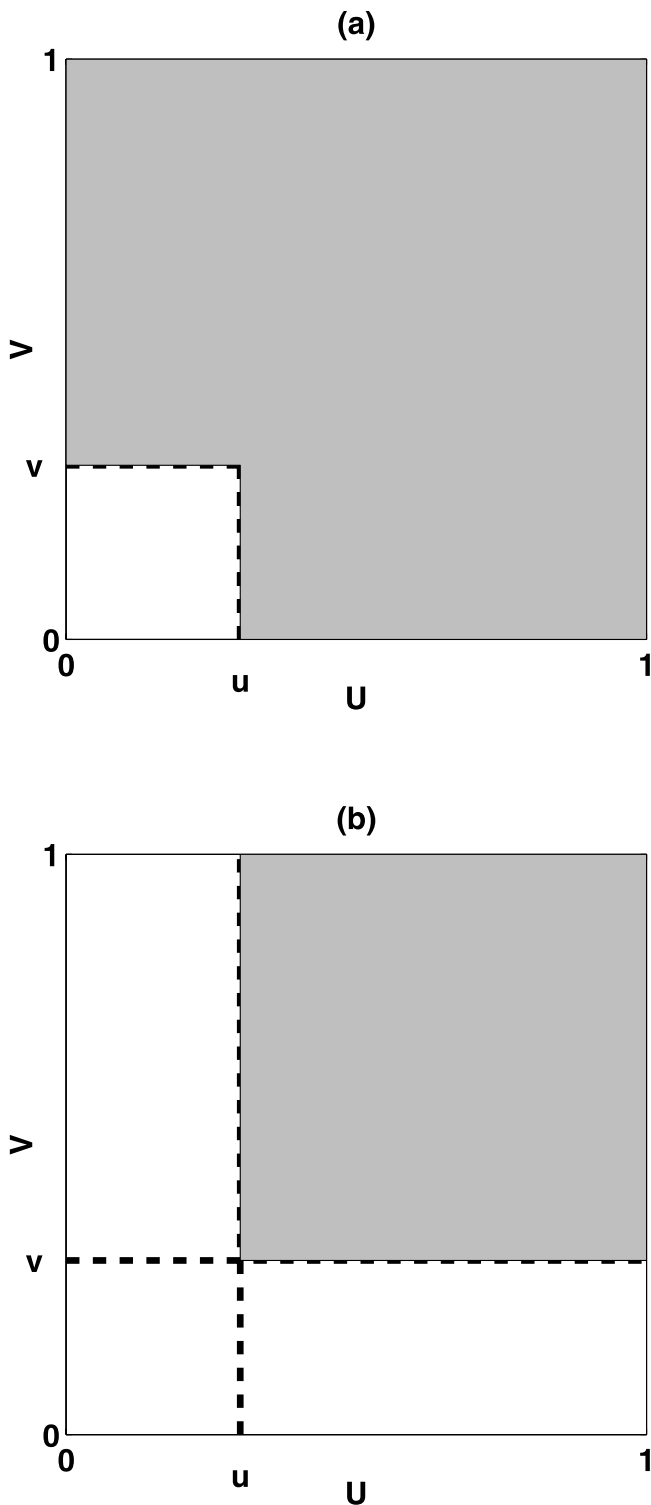


Figure 1. Regions of $\mathbf{I}^2$ (shaded areas) carrying the probability masses given by (a) $p_{u,v}^\vee = \mathbf{P}\{U > u \vee V > v\} = 1 - C(u, v)$ and (b) $p_{u,v}^\wedge = \mathbf{P}\{U > u \wedge V > v\} = 1 - u - v + C(u, v)$.

[17] Hereinafter, we shall consider (X, Y) as the pair of RVs effectively describing the hydrological phenomenon of interest; one of the relevant results of this work will be to show how to make inferences on the behavior of (X, Y) by considering, instead, the pair (U, V) , a trick which may lead

to simpler calculations. Indeed, the fact that the marginals of U and V are uniform laws (and hence independent of any parameter) is of fundamental importance because this turns the original problem into a nonparametric (distribution-free) one, which could be less difficult to solve.

[18] Incidentally, we point out that the study of conditional distributions is greatly facilitated by using copulas. For instance, the following conditional laws (see also section 4.3) can easily be calculated simply by differentiating the corresponding copula [Nelsen, 1999]:

$$\mathbf{P}\{V \leq v \mid U = u\} = \frac{\partial}{\partial u} C(u, v), \quad (3a)$$

$$\mathbf{P}\{U \leq u \mid V = v\} = \frac{\partial}{\partial v} C(u, v). \quad (3b)$$

[19] Note that the above formulas are the mathematical core for simulating copulas. We need now to introduce the following function (Definition 2).

[20] • Let C be a copula, and let $\hat{C}: \mathbf{I}^2 \rightarrow \mathbf{I}$ be the bivariate function given by

$$\hat{C}(u, v) = u + v - 1 + C(1 - u, 1 - v). \quad (4)$$

Then, $\hat{C}$ is a copula called the survival copula of U and V .

[21] Defining the marginal survival functions $\bar{F}_U(u) = \mathbf{P}\{U > u\} = 1 - u$ and $\bar{F}_V(v) = \mathbf{P}\{V > v\} = 1 - v$, it turns out that the joint survival function $\bar{F}_{UV}$ is given by

$$\bar{F}_{UV}(u, v) = \mathbf{P}\{U > u, V > v\} = \hat{C}(\bar{F}_U(u), \bar{F}_V(v)); \quad (5)$$

see Figure 1 for a graphical illustration.

[22] Later, we shall mainly be concerned with a particular subclass of copulas, i.e., those called Archimedean, featuring useful properties. First of all, we need the following Definition 3.

[23] • Let $\gamma: \mathbf{I} \rightarrow [0, \infty]$ such that γ is continuous and strictly decreasing, with $\gamma(1) = 0$. If $\gamma^{(-1)}$ denotes the ordinary inverse function of γ , then the pseudo-inverse of γ is the function $\gamma^{[-1]}: [0, \infty] \rightarrow \mathbf{I}$ given by

$$\gamma^{[-1]}(t) = \begin{cases} \gamma^{(-1)}(t), & 0 \leq t \leq \gamma(0) \\ 0, & \gamma(0) \leq t \leq \infty \end{cases}. \quad (6)$$

[24] Note that $\gamma^{[-1]}$ is continuous and nonincreasing on $[0, \infty]$ and strictly decreasing on $[0, \gamma(0)]$. Furthermore, $\gamma^{[-1]}(\gamma(t)) = t$ on $\mathbf{I}$ and $\gamma(\gamma^{[-1]}(t)) = \min\{t, \gamma(0)\}$; clearly, if $\gamma(0) = \infty$, then $\gamma^{[-1]} = \gamma^{(-1)}$. The following Result 2 implicitly defines the Archimedean copulas.

[25] • Let $C: \mathbf{I} \times \mathbf{I} \rightarrow \mathbf{I}$ be given by

$$C(u, v) = \gamma^{[-1]}(\gamma(u) + \gamma(v)), \quad (7)$$

where $\gamma, \gamma^{[-1]}$ are as in Definition 3. Then, C is a copula if and only if γ is convex.

[26] Note that γ is convex if and only if $\gamma^{[-1]}$ is convex. Thus we have the following Definition 4.

[27] • Copulas given by Result 2 are called Archimedean, with (additive) generator γ . If $\gamma(0) = \infty$, then γ as well as the corresponding copula are called strict.

[28] Archimedean copulas are (1) symmetric (i.e., for all $u, v \in \mathbf{I}$, $\mathbf{C}(u, v) = \mathbf{C}(v, u)$) and (2) associative (i.e., for all $u, v, w \in \mathbf{I}$, $\mathbf{C}(\mathbf{C}(u, v), w) = \mathbf{C}(u, \mathbf{C}(v, w))$); note that the symmetry condition 1 can be relaxed [Joe, 1997].

[29] As an illustration, below we report the analytical expressions of two well-known families of strict Archimedean copulas, together with the corresponding generators, which we shall use later in section 4.

[30] The standard expression for the members of the Gumbel-Hougaard family is [Hutchinson and Lai, 1990; Nelsen, 1999]

$$\mathbf{C}(u, v) = e^{-[(-\ln u)^\delta + (-\ln v)^\delta]^{1/\delta}}, \quad (8)$$

where $u, v \in \mathbf{I}$ and $\delta \geq 1$ is a dependence parameter. In particular, U and V are independent for $\delta = 1$ and are positively associated for $\delta > 1$ (i.e., small values of U are associated with small values of V , and large values of U are associated with large values of V); further mathematical properties can be found by Nelsen [1999]. The expression of the corresponding generator is simple and is given by, for $t \in \mathbf{I}$,

$$\gamma(t) = (-\ln t)^\delta. \quad (9)$$

[31] One of the possible equivalent expressions for the members of the Frank family is [Frank, 1979; Nelsen, 1986; Genest, 1987]

$$\mathbf{C}(u, v) = \frac{1}{\ln \delta} \ln \left(1 + \frac{(\delta^u - 1)(\delta^v - 1)}{\delta - 1} \right), \quad (10)$$

where $u, v \in \mathbf{I}$ and $\delta > 0$ is a dependence parameter. The RVs (U, V) are positively associated for $0 < \delta < 1$ and are negatively associated for $\delta > 1$. The case $\delta = 1$ corresponds to independent RVs. Further mathematical properties can be found by Nelsen [1999]. The expression of the corresponding generator is simple and is given by, for $t \in \mathbf{I}$,

$$\gamma(t) = -\ln \left(\frac{\delta^t - 1}{\delta - 1} \right). \quad (11)$$

These are the only Archimedean copulas which satisfy the functional equation $\mathbf{C} = \hat{\mathbf{C}}$ for radial symmetry; as a consequence, also $\hat{\mathbf{C}}$ is Archimedean, and the two copulas have the same generator γ .

[32] Although Archimedean copulas do not embrace the whole family of copulas, nevertheless, they may suffice for modeling quite a few phenomena (see section 4). In particular, they will play a fundamental role in the following, essentially because their level curves (sets) can be written explicitly (Definition 5), as follows.

[33] • Given $0 < t \leq 1$, the curves

$$L_t = \{(u, v) \in \mathbf{I}^2: \mathbf{C}(u, v) = t\} \quad (12)$$

are called the level curves of $\mathbf{C}$.

[34] The line L_t connects the border points $(t, 1)$ and $(1, t)$ since $\mathbf{C}(t, 1) = \mathbf{C}(1, t) = t$ by virtue of Definition 1. Instead, for $t = 0$, $Z(\mathbf{C}) = L_0$ represents the zero set of $\mathbf{C}$, i.e., the region of $\mathbf{I}^2$ where $\mathbf{C}(u, v) = 0$ or, equivalently, the region of

the impossible events; clearly, $Z(\mathbf{C})$ may either (1) reduce to the two line segments $\{0\} \times \mathbf{I}$ and $\mathbf{I} \times \{0\}$ or (2) have a positive area.

[35] Now, let $0 < t \leq 1$. In the Archimedean case, L_t is convex. Also, since $\mathbf{C}$ is constant on L_t by virtue of equation (12), thanks to equation (7), it is clear that the points of L_t satisfy the relation

$$(u, v) \in L_t \iff \gamma(u) + \gamma(v) = \gamma(t). \quad (13)$$

Therefore it is possible to express v as a function of u (and vice versa) for $t \leq u \leq 1$, simply calculating $v = L_t(u)$, where

$$L_t(u) = \gamma^{-1}(\gamma(t) - \gamma(u)) = \gamma^{(-1)}(\gamma(t) - \gamma(u)). \quad (14)$$

Note that replacing $\gamma^{[-1]}$ by $\gamma^{(-1)}$ is justified since $\gamma(t) - \gamma(u)$ is in the interval $[0, \gamma(0)]$.

3. Bivariate Return Periods

[36] In this section we introduce and discuss the concept of return period for events defined by the joint behavior of pairs of RVs. As will be seen, several definitions are possible, each emphasizing particular facets of the dynamics of the phenomenon considered. Below, we shall skip all the unnecessary mathematical details; a thorough treatment can be found by Salvadori [2004].

[37] Let us consider a sequence $E_1, E_2, \dots$ of independent events; for the sake of definiteness we may assume that such events happen in the temporal domain at times $t_1 < t_2 < \dots$ (i.e., we use a temporal marked point process as a model). Each event E_i is characterized by the joint behavior of a pair of RVs $(U, V) \sim \mathbf{C}$, which can be expressed in terms of the following set of marginal events:

$$E_{U,u}^< = \{U \leq u\}, \quad (15a)$$

$$E_{U,u}^> = \{U > u\}, \quad (15b)$$

$$E_{V,v}^< = \{V \leq v\}, \quad (15c)$$

$$E_{V,v}^> = \{V > v\}. \quad (15d)$$

Clearly, using the (inclusive) OR operator “ $\vee$ ” and the AND operator “ $\wedge$,” it is possible to combine the marginal events in several ways, as shown in Table 1.

[38] Hydrologic systems are impacted by extreme events, which may produce huge damage to natural environments and human lives. However, the severity of phenomena such as droughts and floods may be the consequence of a different behavior of the same cause, i.e., the precipitation: too small for too long a time in the former case, too large in the latter. Actually, the eight combinations shown in Table 1 summarize all the possible different types of bivariate events E_i of interest.

[39] Although the theoretical investigations outlined in the following could easily be generalized to all the eight types of joint events shown in Table 1, for the sake of

Table 1. Possible Joint Events Using the (Inclusive) OR “ $\vee$ ” and the AND “ $\wedge$ ” Operators

	$E_{U,u}^{\leq}$	$E_{U,u}^{>}$
	$\vee$	
$E_{V,v}^{\leq}$	$\{U \leq u\} \vee \{V \leq v\}$	$\{U > u\} \vee \{V \leq v\}$
$E_{V,v}^{>}$	$\{U \leq u\} \vee \{V > v\}$	$\{U > u\} \vee \{V > v\}$
	$\wedge$	
$E_{V,v}^{\leq}$	$\{U \leq u\} \wedge \{V \leq v\}$	$\{U > u\} \wedge \{V \leq v\}$
$E_{V,v}^{>}$	$\{U \leq u\} \wedge \{V > v\}$	$\{U > u\} \wedge \{V > v\}$

brevity, in the remaining part of the work we shall mainly be concerned with the following extremal events:

$$E_{u,v}^{\vee} = \{U > u\} \vee \{V > v\} \quad (16a)$$

$$E_{u,v}^{\wedge} = \{U > u\} \wedge \{V > v\}. \quad (16b)$$

Indeed, in some applications, $E_{u,v}^{\vee}$ and $E_{u,v}^{\wedge}$ are the ones of greatest interest. In fact, in practice, an event could be defined as “dangerous” if (1) either U or V exceed given thresholds (in which case $E_{u,v}^{\vee}$ is the relevant event) or (2) both U and V are larger than prescribed values (in which case $E_{u,v}^{\wedge}$ is the relevant event). For instance, a storm may be considered as dangerous if either its (average) intensity is too large or its duration is too long; as a further example, if both flood-volume and flood-peak are too high, then a dam safety may be at risk.

[40] Now, let T be the interarrival time between two successive events E_i and E_{i+1} , $i = 1, 2, \dots$, with $\mu_T = E(T) < \infty$. As is natural, we assume that $T > 0$ almost surely, and therefore $\mu_T > 0$. For instance, T may represent the time between the arrival of two successive rainfall storms in a given region; as a further practical example, if the observations considered are annual values, then $\mu_T = 1$ year. Assuming that the interarrival times are independent and identically distributed (and independent of U and V), via the Wald equation [see, e.g., Ross, 1996], it is easy to show that

$$\tau_{u,v}^{\vee} = E(T_{u,v}^{\vee}) = \mu_T / p_{u,v}^{\vee} \quad (17a)$$

$$\tau_{u,v}^{\wedge} = E(T_{u,v}^{\wedge}) = \mu_T / p_{u,v}^{\wedge}, \quad (17b)$$

where $T_{u,v}^{\vee}$ denotes the interarrival time between two successive events of type $E_{u,v}^{\vee}$; $p_{u,v}^{\vee}$ represents the probability of the event $E_{u,v}^{\vee}$; $T_{u,v}^{\wedge}$ denotes the interarrival time between two successive events of type $E_{u,v}^{\wedge}$; and $p_{u,v}^{\wedge}$ represents the probability of the event $E_{u,v}^{\wedge}$. We thus have the following Definition 6.

[41] • The positive numbers $\tau_{u,v}^{\vee}$ and $\tau_{u,v}^{\wedge}$ given by equation (17) define, respectively, the return periods of the events $E_{u,v}^{\vee}$ and $E_{u,v}^{\wedge}$.

[42] Note that $\tau_{u,v}^{\vee}$, $\tau_{u,v}^{\wedge}$ are decreasing functions of the corresponding probabilities $p_{u,v}^{\vee}$, $p_{u,v}^{\wedge}$. This is obvious since the interarrival time gets longer for less probable events. Using copulas, it is easy to calculate the latter probabilities:

$$p_{u,v}^{\vee} = \mathbf{P}\{U > u \vee V > v\} = 1 - \mathbf{C}(u, v) \quad (18a)$$

$$p_{u,v}^{\wedge} = \mathbf{P}\{U > u \wedge V > v\} = \hat{\mathbf{C}}(1 - u, 1 - v). \quad (18b)$$

The pictures in Figure 1 illustrate the regions of $\mathbf{I}^2$ carrying the probability masses given by $p_{u,v}^{\vee}$ and $p_{u,v}^{\wedge}$. Evidently, since $\mathbf{C}(u, v)$ is nonnegative and nondecreasing in u and v ,

$$p_{u,v}^{\vee} \geq p_{u,v}^{\wedge} \quad \text{and} \quad \tau_{u,v}^{\vee} \leq \tau_{u,v}^{\wedge}. \quad (19)$$

Indeed, it is sufficient that either $E_{u,u}^{\vee}$ or $E_{v,v}^{\vee}$ happens for the realization of $E_{u,v}^{\vee}$, whereas it is necessary that both $E_{u,u}^{\vee}$ and $E_{v,v}^{\vee}$ happen for the realization of $E_{u,v}^{\wedge}$. In addition, more probable events have shorter return periods.

[43] Now, consider the invertible transformations (see also equation (2) and the ensuing discussion)

$$\begin{cases} u = F_X(x) \\ v = F_Y(y) \end{cases} \iff \begin{cases} x = F_X^{(-1)}(u) \\ y = F_Y^{(-1)}(v) \end{cases}, \quad (20)$$

where $(u, v) \in \mathbf{I}^2$ and $(x, y) \in \mathbf{R}^2$. Clearly, for fixed (x, y) , by using equation (17), (18), and (20), it is immediate to calculate $\tau_{x,y}^{\vee}$ and $\tau_{x,y}^{\wedge}$ and hence the return periods of the events $E_{x,y}^{\vee}$ and $E_{x,y}^{\wedge}$ as expressed in terms of the pair (X, Y) .

3.1. Bivariate Versus Univariate Analysis

[44] As thoroughly illustrated by Yue and Rasmussen [2002], if a given hydrological event is multivariate, i.e., described by a set of nonindependent RVs, then univariate frequency analysis cannot provide complete assessment of the probability of occurrence. Actually, a better understanding of the statistical characteristics of such events requires the study of their joint distribution.

[45] Thus a first important consideration is motivated by a well-known hydrological problem, which we briefly illustrate in the following (see also De Michele et al. [2004] and section 4.2): the determination of the spillway design flood. Its calculation is commonly based on the univariate quantiles x_{π} , y_{π} of the maximum annual flood volume X and the maximum annual flood peak Y , respectively, for the same return period π . Usually, $\pi = 1000$ years (sometimes $\pi = 5000$ or $10,000$ years, depending upon the magnitude of damages that a failure could create, and the specific directives followed in each country). Note that the assumption of a common return period π is used in the present context only.

[46] Let us first make some issues clear. In practical applications, what matters above all is not the univariate, bivariate, or even multivariate approach but the definition of risk and the choice of the variable(s) to study. Indeed, the potential over/underestimation in the design may be due to an inappropriate choice of the variable studied (e.g., peak of inflow hydrograph rather than peak of outflow hydrograph) and not to the univariate approach per se [see, e.g., Koutsoyiannis, 1994].

[47] The univariate frequency analysis is useful when one RV is significant in the design process. For instance, if the outflow peak is the variable studied in a reservoir spillway design, then the univariate approach is fine. Now, the flood peak is an essential design factor for a reservoir spillway; however, if the reservoir capacity is large, then the flood

Table 2. Differences Between Univariate and Bivariate Estimates for the Case Study Illustrated in Section 4.2^a

π , years	z_π	$\tau_{z_\pi, z_\pi}^\vee$, years	$\tau_{z_\pi, z_\pi}^\wedge$, years	$z_\pi^\vee$	$z_\pi^\wedge$
10	0.9	8.1	13.1	0.91946	0.86960
100	0.99	79.8	133.9	0.99202	0.98662
1000	0.999	797.1	1341.4	0.99920	0.99866
10,000	0.9999	7970.2	13,417.0	0.99992	0.99987

^aSee text.

volume is also an important safety factor in the reservoir design for two reasons. First, even if the reservoir is full at the beginning of the flood, there will be some temporary storage during the flood routing through the spillway; this may reduce significantly the outflow peak if the flood volume is low or may not reduce it if the flood volume is high. Second, the probability that the reservoir has some empty space in the beginning of the flood to store part of the flood volume may be significant. This also may affect significantly the outflow peak, which obviously is the factor that determines the risk. As a consequence, the bivariate analysis of the return periods of flood volume and flood peak may provide more useful information for design criterion than a univariate investigation. In addition, the (generalized) bivariate approach we propose may be useful not only in the design of structures but in many other practical situations too.

[48] Thus let us suppose that $(X, Y) \sim \mathbf{C}$ and, in particular, that X and Y are not independent (or just weakly dependent). Then, introduce the quantiles $u_\pi = F_X(x_\pi)$ and $v_\pi = F_Y(y_\pi)$ for the corresponding pair (U, V) . Clearly, since π is the same for both X and Y , then $u_\pi = v_\pi$; thus let z_π denote their common value. Now, since copulas are nondecreasing in u and v , it is immediate to realize that their diagonal section $\delta(z) = \mathbf{C}(z, z)$ satisfies the inequality $\delta(z) \leq z$ for all $z \in \mathbf{I}$ (if $\mathbf{C}$ is Archimedean, then the inequality is strict for $z \in (0, 1)$). In turn,

$$\tau_{z_\pi, z_\pi}^\vee \leq \pi \leq \tau_{z_\pi, z_\pi}^\wedge. \quad (21)$$

As a consequence, $E_{z_\pi, z_\pi}^\vee$ and $E_{z_\pi, z_\pi}^\wedge$ might not be π years bivariate events. More specifically, if $E_{z_\pi, z_\pi}^\vee$ were used as a “critical” design event, in order to have $\tau_{z_\pi, z_\pi}^\vee = \pi$, the marginal quantiles x_π, y_π (and hence z_π) might need to be increased. By the same token, if $E_{z_\pi, z_\pi}^\wedge$ were used as a “critical” design event, in order to have $\tau_{z_\pi, z_\pi}^\wedge = \pi$, the marginal quantiles x_π, y_π (and hence z_π) might need to be decreased. In other words, if x_π, y_π were used as design quantiles, then the resulting work could be either underdimensioned (and would be at risk) or overdimensioned (yielding a waste of money).

[49] Using the values of the parameters estimated by *De Michele et al.* [2004] (see also section 4.2 below), in Table 2 we calculate the return periods $\tau_{z_\pi, z_\pi}^\vee$ (for $E_{z_\pi, z_\pi}^\vee$) and $\tau_{z_\pi, z_\pi}^\wedge$ (for $E_{z_\pi, z_\pi}^\wedge$) for several choices of π . Now, suppose that the desired return period is π . Evidently, as prescribed by the inequality in equation (21), the return period of the event $E_{z_\pi, z_\pi}^\vee$ is about 20% smaller than π , while the return period of the event $E_{z_\pi, z_\pi}^\wedge$ is about 30% larger than π . As already pointed out, in order to obtain joint events having a return period equal to π , it is necessary to consider values of the

quantiles different from z_π . The last two columns in Table 2 show which quantiles (respectively, $z_\pi^\vee$ and $z_\pi^\wedge$) should be used for obtaining events $E_{z_\pi^\vee, z_\pi^\vee}^\vee$ and $E_{z_\pi^\wedge, z_\pi^\wedge}^\wedge$ having a return period equal to π . Note that the correct values of the quantiles always satisfy the inequality

$$z_\pi^\wedge \leq z_\pi \leq z_\pi^\vee, \quad (22)$$

as anticipated while discussing equation (21).

[50] As a first conclusion, the analysis of bivariate return periods is greatly facilitated by using copulas. They may give the hydrologists a precise understanding of the stochastic joint dependence of the variables of interest as well as the possibility of correctly sizing the works, which would be hard to achieve adopting a simple univariate approach.

[51] A further practical problem, of the highest interest in the applications, is the identification of the events having an assigned return period. In general, its solution is not unique. In fact, it may happen that different choices of u and v (or, equivalently, different realizations of X and Y) yield the same return period. Thus the problem is to identify in the unit square $\mathbf{I}^2$ (or, equivalently, in the plane $\mathbf{R}^2$) the isolines of $\tau_{u,v}^\vee$ and $\tau_{u,v}^\wedge$, which correspond to curves of isofrequency for the events considered. In the next sections we shall deal with the problem of calculating the isolines of $\tau_{u,v}^\vee$ and $\tau_{u,v}^\wedge$.

3.2. Calculating the Isolines of $\tau_{u,v}^\vee$

[52] Since $\tau_{u,v}^\vee$ is constant in $\mathbf{I}^2$ wherever $p_{u,v}^\vee$ is, in order to find the isolines of $\tau_{u,v}^\vee$, it is enough to search for those of $\mathbf{C}$. In general, the task is nontrivial from an analytical point of view and may require the use of numerical routines. However, if we assume that $\mathbf{C}$ is Archimedean, then we obtain, for $0 < t \leq 1$,

$$p_{u, L_t(u)}^\vee = 1 - \mathbf{C}(u, L_t(u)) = 1 - t, \quad t \leq u \leq 1, \quad (23)$$

i.e., $p_{u,v}^\vee$ is constant on the (convex) level curve $v = L_t(u)$, which can be written explicitly using equation (14). As a consequence,

$$\tau_{u, L_t(u)}^\vee = \vartheta(t) = \frac{\mu_T}{1-t}, \quad (24)$$

where the function $\vartheta: \mathbf{I} \rightarrow [\mu_T, \infty]$ is one-to-one and increasing. Thus the events $E_{u, L_t(u)}^\vee$ have a constant return period. In turn, the return period of the corresponding events for the pair (X, Y) is constant in $\mathbf{R}^2$ on the set

$$\begin{cases} x = F_X^{(-1)}(u) \\ y = F_Y^{(-1)}(L_t(u)) \end{cases}, \quad (25)$$

where $x \geq F_X^{(-1)}(t)$. Note that $p_{u, L_t(u)}^\vee$ reduces by increasing t , while the corresponding return period $\tau_{u, L_t(u)}^\vee$ increases. In fact, increasing t is equivalent to choose more “extreme” events, which have longer interarrival times.

[53] A further important point is the following. Let

$$B_C(t) = \{(u, v) \in \mathbf{I}^2: \mathbf{C}(u, v) \leq t\}, \quad 0 < t \leq 1. \quad (26)$$

Evidently, $B_C(t)$ is the region in $\mathbf{I}^2$ lying on, or below and to the left of, the level curve L_t . Since $\mathbf{C}$ induces in a natural way a unique probability measure on $\mathbf{I}^2$, it is possible to use

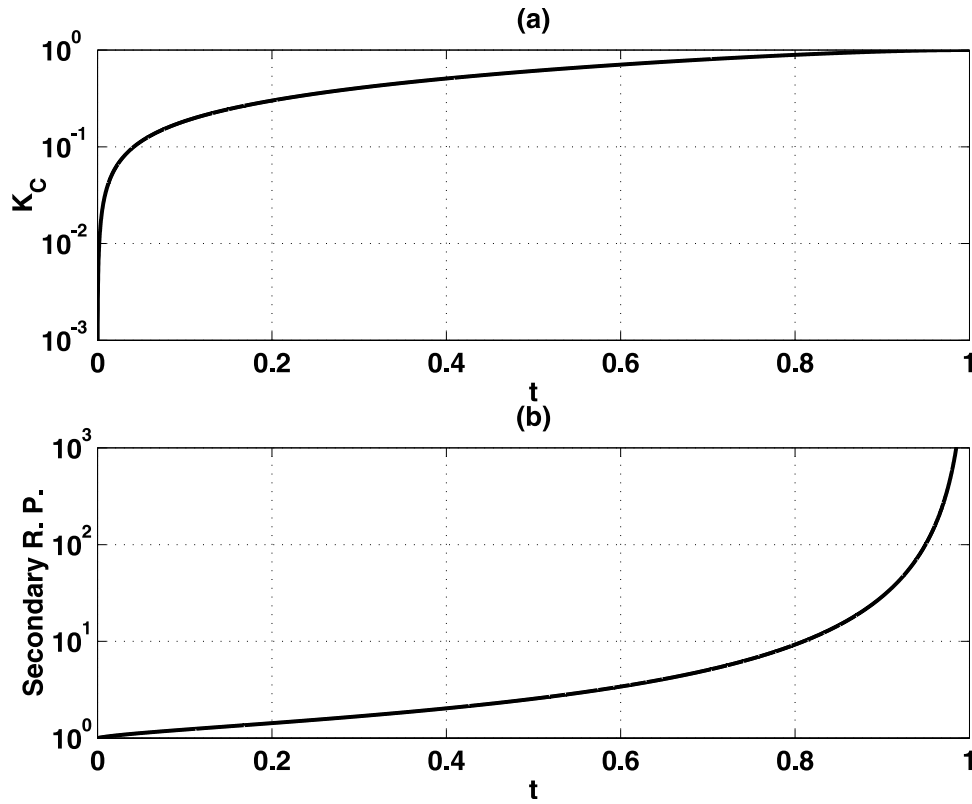


Figure 2. Illustration of (a) the probability function $K_C(t)$ and (b) the secondary return period function Q_t^V .

this fact in order to introduce a suitable measure of the sets B_C , as stated by the following Result 3 (for all the mathematical details, see Nelsen [1999]).

[54] • Let $0 < t \leq 1$. The function $K_C(t)$ given by

$$K_C(t) = t - \frac{\gamma(t)}{\gamma'(t^+)} \quad (27)$$

provides a unique probability measure of the set $B_C(t)$.

[55] Here $\gamma'(t^+)$ represents the right-sided derivative of γ , which exists almost everywhere in $\mathbf{I}$. It is now fundamental to consider the following equivalence, where $0 < t \leq 1$:

$$(u, v) \in B_C(t) \iff \tau_{u,v}^V \leq \vartheta(t). \quad (28)$$

In other words, $B_C(t)$ is exactly the region in $\mathbf{I}^2$ where the return period $\tau_{u,v}^V$ is less than or equal to a well-defined monotone function of t . Thus $B_C(t)$ induces a partition on the family of events $\mathcal{E}^V = \{E_{u,v}^V\}$, $(u, v) \in \mathbf{I}^2$: those belonging to $B_C(t)$ (having return period $\tau_{u,v}^V \leq \vartheta(t)$) and those not belonging to $B_C(t)$ (having return period $\tau_{u,v}^V > \vartheta(t)$). In addition, every threshold $\vartheta(t)$ can be uniquely associated with a probability $K_C(t)$. Thus given $0 < t < 1$, the (survival) function

$$\bar{K}_C(t) = 1 - K_C(t), \quad (29)$$

yields the probability that at any given realization of the process (e.g., at any storm), an event $E_{u,v}^V \in \mathcal{E}^V$ having return period $\tau_{u,v}^V > \vartheta(t)$ will show up. Clearly, such events are the (possibly) destructive ones; see the discussion below. The

interesting point is that it is then possible to introduce a new variable Q_t^V as follows:

$$Q_t^V = \frac{\mu_T}{1 - K_C(t)} = \vartheta(K_C(t)), \quad (30)$$

where $0 < t < 1$. This yields the following Definition 7.

[56] • Let $0 < t < 1$. The positive number Q_t^V given by equation (30) is called the secondary return period of the events outside $B_C(t)$.

[57] The label “secondary” is chosen to emphasize the difference with $\tau_{u,v}^V$, which could well be named as “primary.” Note that Q_t^V features all the properties of a true return period; in particular, it concentrates into a single number the relevant information about the realization of dangerous events. As a matter of fact, equation (30) precisely states that Q_t^V satisfies (formally and practically) the conditions for being defined as a return period. As an illustration, in Figure 2 we plot $K_C(t)$ and Q_t^V as functions of $t \in \mathbf{I}$. Here C and the relevant parameters are as in section 4.3; the vertical scale is logarithmic in order to enhance the behavior of the functions plotted.

[58] The above results provide useful information for the design process. However, the new concepts introduced deserve further discussion. In fact, suppose that a critical design threshold $\tau_*^V > \mu_T$ (or, equivalently, a critical level $t_* > 0$) is fixed: for instance, the return period τ_*^V could be the one chosen for sizing a given manufacture. Then, let us consider the unit square $\mathbf{I}^2$, as shown in Figure 3. The continuous line connecting the points $(t_*, 1)$ and $(1, t_*)$ is the isoline of the events $E_{u,v}^V$ having a constant primary return period $\tau_{u,v}^V = \tau_*^V = \vartheta(t_*)$; one of such events is E_*

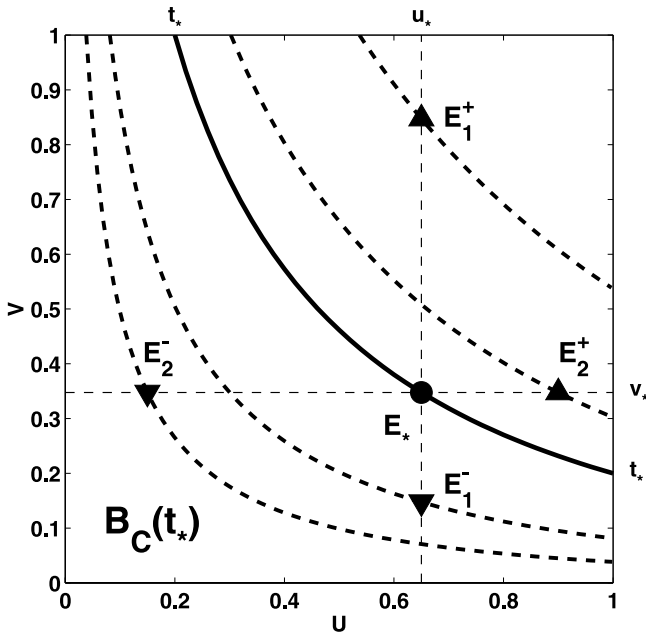


Figure 3. Illustration of sub-, super-, and critical events (see text).

(circle); clearly, the points of this level curve satisfy the equation $v = L_{t_*}(u)$, which can be written explicitly. As a consequence, $B_C(t_*)$ is the region in $\mathbf{I}^2$ lying on, or below and to the left of, the convex level curve L_{t_*} . As already mentioned, all the events in $B_C(t_*)$ have primary return period $\tau_{u,v}^\vee \leq \vartheta(t_*)$. In particular, those events on L_{t_*} are the critical ones (having primary return period $\tau_{u,v}^\vee = \tau_*^\vee$), while the others are subcritical. In Figure 3, two different subcritical events E_1^- , E_2^- are shown, plotted as downward triangles. Note that the primary return period of E_2^- is smaller than the one of E_1^- . In addition, all the events on the level curve crossing E_1^- (dashed line) share the same primary return period as of E_1^- , while all the events on the level curve crossing E_2^- share the same primary return period as of E_2^- . Finally, all the events outside $B_C(t_*)$ have primary return period $\tau_{u,v}^\vee > \tau_*^\vee$ and represent the supercritical ones. In Figure 3, two different supercritical events E_1^+ , E_2^+ are shown, plotted as upward triangles. Note that the primary return period of E_2^+ is smaller than the one of E_1^+ . In addition, all the events on the level curve crossing E_1^+ share the same primary return period as of E_1^+ , while all the events on the level curve crossing E_2^+ share the same primary return period as of E_2^+ .

[59] A further fundamental consideration is as follows. The primary return period, the one usually adopted in the applications for sizing the manufactures, may only provide partial and vague information about the realization of the events of interest. In fact, it only predicts that a critical event is expected to appear once in a given time interval (i.e., an average forecast). However, it would be more important to be able to calculate (1) the probability that a supercritical (destructive) event will show up at any given realization of the process (e.g., at any storm) and (2) how long will it take, on the average, for a supercritical event to appear. As a fundamental result, both questions can now easily be

answered: the first one by using equation (29) and the second one by considering the secondary return period given by equation (30). Thus the secondary return period provides precise indications for performing risk analysis and may also yield useful hints for doing numerical simulations.

[60] From a practical point of view the introduction of the function K_C turns the difficult analysis of the bivariate dynamics of X and Y into a more simple one-dimensional problem. In fact, at any given realization of the process, only two mutually exclusive things may happen: either the event is supercritical, and hence potentially dangerous, or not. Actually, this is the only thing that matters to the engineers. Thus in the applications the investigation of the joint behavior of X and Y could be reduced to the analysis of the associated (univariate) Bernoulli process, in which the “success” is simply the realization of a supercritical event. As explained above, given a critical design level t_* , all the supercritical events are enclosed in a sort of “equivalence class” (practically, the region outside $B_C(t_*)$), whose probability can easily be calculated as $\bar{K}_C(t_*)$ via equation (29). Thus whatever the values of X and Y occurring in the phenomenon of interest, it is now possible to discriminate between sub- and supercritical events, judge whether an event is supercritical or not by precisely indicating which events are supercritical, calculate their exact probability at any given realization of the process, and even estimate their average rate of recurrence via the secondary return period.

[61] A final point concerns the study of the (marginal) quantiles of the events of interest. As before, assume that a critical design level t_* is fixed, and let L_{t_*} be the isoline of the critical events. Clearly, the coordinates (u, v) of any subcritical event $E_{u,v}^\vee$ must satisfy the inequality $v < L_{t_*}(u)$ since the event must lie below and to the left of L_{t_*} . This is well evident in Figure 3 considering the subcritical event E_1^- . Conversely, the coordinates of a supercritical event must satisfy the inequality $v > L_{t_*}(u)$ since the event must lie above and to the right of L_{t_*} . Again, this is well evident in Figure 3 considering the supercritical event E_1^+ . Equivalently, in terms of the RVs (X, Y) of practical interest, and considering a subcritical event, for any arbitrary choice of the X s marginal quantile

$$x_q > x_* = F_X^{(-1)}(t_*),$$

the corresponding Y s marginal quantile y_q must satisfy the inequality

$$y_q < y_q^* = F_Y^{(-1)}(L_{t_*}(F_X(x_q)))$$

and the inequality $y_q > y_q^*$ when considering a supercritical event. Thus a fundamental information is achieved: if $x_q > x_*$ is already given by design requirements, and a (joint) primary return period $\tau_*^\vee$ is desired, then all the events with $Y > y_q^*$ must be considered as supercritical and hence as potentially dangerous. The same reasoning can be applied to X , simply by swapping the roles of X and Y . In particular, note that the three events E_1^- , E_* , E_1^+ have the same X s marginal quantile $x_* = F_X^{(-1)}(u_*)$, but the corresponding Y s marginal quantiles satisfy the inequality $y_- < y_* = F_Y^{(-1)}(v_*) < y_+$. Indeed, these events are, subcritical, critical, and supercritical, respectively, with

respect to the threshold t_* . The same comments hold for the three events E_2^- , E_* , E_2^+ , swapping the roles of X and Y .

3.3. Calculating the Isolines of $\tau_{u,v}^\wedge$

[62] Essentially, finding the isolines of constant return period for the event $E_{u,v}$ requires the same techniques used in section 3.2, with the obvious changes. In general, the task is nontrivial from an analytical point of view and may require the use of numerical routines.

[63] The starting point of our analysis consists in the observation that the following relations are equivalent:

$$\begin{aligned} p_{u,v}^\wedge = 1 - t &\iff \hat{\mathbf{C}}(1 - u, 1 - v) = 1 - t \\ &\iff u + v - \mathbf{C}(u, v) = t, \end{aligned} \quad (31)$$

where $0 < t \leq 1$. Thus, as before, if $\mathbf{C}$ or $\hat{\mathbf{C}}$ were Archimedean, then we could hope for an easy solution.

[64] Now, let $\hat{\mathbf{C}}$ be Archimedean with generator γ , and consider the second equality in equation (31). Then, the level curve $v = L_t(u)$ given by

$$v = L_t(u) = 1 - \gamma^{(-1)}(\gamma(1 - t) - \gamma(1 - u)), \quad (32)$$

with $0 \leq u \leq t$, yields the set in $\mathbf{I}^2$, where $p_{u,v}^\wedge$ (and, consequently, $\tau_{u,v}^\wedge$) is constant. In particular,

$$\tau_{u, L_t(u)}^\wedge = \vartheta(t). \quad (33)$$

[65] Alternatively, consider again equation (31) and suppose that $\mathbf{C}$ is Archimedean, with generator γ . Introducing the new variables a, b given by

$$\begin{cases} a = \gamma(u) \\ b = \gamma(v) \end{cases} \iff \begin{cases} u = \gamma^{[-1]}(a) \\ v = \gamma^{[-1]}(b) \end{cases}, \quad (34)$$

the last equality in equation (31) can be written in a more tractable form which only involves the monotone function $\gamma^{[-1]}$, i.e.,

$$\gamma^{[-1]}(a) + \gamma^{[-1]}(b) - \gamma^{[-1]}(a + b) = t. \quad (35)$$

[66] In general, the pairs (a, b) satisfying equation (35) cannot be determined analytically; however, finding a numerical solution is not too difficult a task. Then, using equations (34) and (20), it is easy to find the isolines of constant return period in $\mathbf{R}^2$ of the event $E_{x,y}^\wedge$, as expressed in terms of the pair (X, Y) . Note that given the symmetries in equation (35) (the roles of a and b are exchangeable), the search for a numerical solution, say, b as a function of a , may start at $a = \gamma(t)$ (yielding $b = \gamma(0)$, including the case $b = +\infty$, and hence the solution $(u = t, v = 0)$) and go on by increasing a up to $a = a'$, calculated via the equation

$$2\gamma^{[-1]}(a') - \gamma^{[-1]}(2a') = t.$$

Further solutions for b can then be found simply by swapping a and b in the results obtained so far.

3.4. Conditional Return Periods

[67] As anticipated when stating equation (3), the study of conditional distributions is greatly facilitated by adopting

copulas. As a further result, using the same notation as above, it is possible to introduce the concept of conditional return period. In fact, in the applications the study of the return period of specific conditional events is also of interest (see section 4); note that conditional events other than the ones considered here could also be investigated using the same approach. With a little abuse of mathematical notation, let us first define the quantities we shall investigate below (Definition 8).

[68] • Let $E_{u,v}^{U|V}$ and $E_{v,u}^{V|U}$ denote the conditional events

$$E_{u,v}^{U|V} = \{U > u \mid V > v\} \quad (36a)$$

$$E_{v,u}^{V|U} = \{V > v \mid U > u\}, \quad (36b)$$

respectively. The corresponding conditional return periods are denoted by $\tau_{u|v}^\wedge$ and $\tau_{v|u}^\wedge$, respectively.

[69] Thus the conditional events $E_{u,v}^{U|V}$ and $E_{v,u}^{V|U}$ can be expressed in terms of the marginal events $E_{u,u}^\wedge$ and $E_{v,v}^\wedge$ given by equations (15b) and (15d): the former concerns the (excess) behavior of U given that $V > v$, while the latter concerns that of V given that $U > u$. Note that the marginal return periods of the univariate events $E_{u,u}^\wedge$ and $E_{v,v}^\wedge$ (denoted by $\tau_{u,u}^\wedge$ and $\tau_{v,v}^\wedge$, respectively) are simply given by

$$\tau_{u,u}^\wedge = \mu_T / \bar{F}_U(u) = \frac{\mu_T}{1 - u} = \vartheta(u) \quad (37a)$$

$$\tau_{v,v}^\wedge = \mu_T / \bar{F}_V(v) = \frac{\mu_T}{1 - v} = \vartheta(v). \quad (37b)$$

[70] Now, let us reconsider the temporal sequence of events $E_1, E_2, \dots$ introduced at the beginning of section 3. Conditioning with respect to the event $E_{v,v}^\wedge = \{V > v\}$ means that we need first to restrict our attention only to those events E_i s where $V > v$ and hence extract from $\{E_i\}$ a well-defined subsequence $\{E_j'\}$; note that the average recurrence time for the events in $\{E_j'\}$ is $\tau_{v,v}^\wedge$. Then, among the events of this latter sequence, we should only consider those in which $U > u$ and thus extract from $\{E_j'\}$ a further well-defined subsequence $\{E_k''\}$. Our target is to calculate the average time elapsing between two successive realizations of the events in $\{E_k''\}$. Clearly, the same approach holds when conditioning with respect to the event $E_{u,u}^\wedge = \{U > u\}$, simply by swapping the roles of U and V .

[71] The elementary reasoning given above yields that in order to calculate $\tau_{u|v}^\wedge$ ($\tau_{v|u}^\wedge$), it is necessary to operate a random Bernoulli selection over the sequence of events $E_{v,v}^\wedge$ s ($E_{u,u}^\wedge$ s), with probability of “success” equal to $\mathbf{P}\{E_{u,u}^\wedge \mid E_{v,v}^\wedge\}$. As a consequence, the conditional return periods are simply given by

$$\tau_{u|v}^\wedge = \tau_{v,v}^\wedge / p_{u,v}^\wedge = \frac{\mu_T}{\bar{F}_V(v)} \frac{1}{\hat{\mathbf{C}}(1 - u, 1 - v)} \quad (38a)$$

$$\tau_{v|u}^\wedge = \tau_{u,u}^\wedge / p_{u,v}^\wedge = \frac{\mu_T}{\bar{F}_U(u)} \frac{1}{\hat{\mathbf{C}}(1 - u, 1 - v)}. \quad (38b)$$

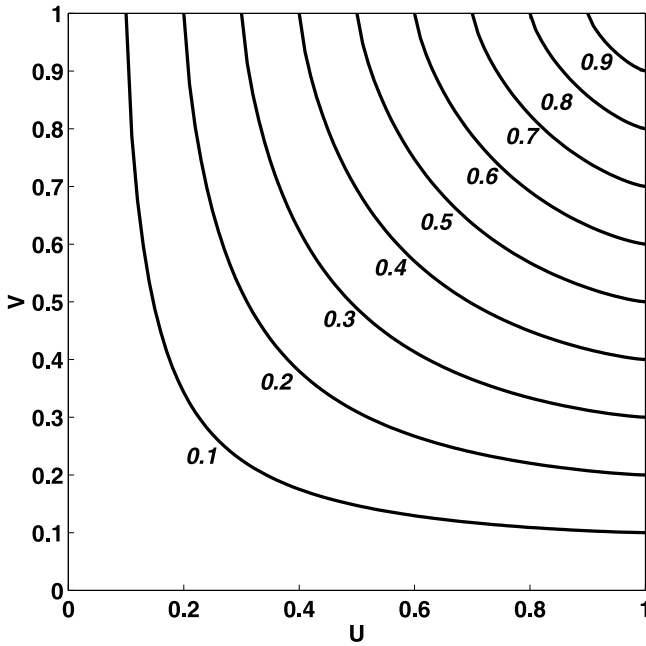


Figure 4. Level curves (isolines) of constant probability of the Gumbel-Hougaard copula for the data considered in section 4.1; the levels t are as indicated.

Thus in order to calculate the isolines of $\tau_{u|v}^{\geq}$, $\tau_{v|u}^{\geq}$, it suffices to solve the equations

$$(1-v)\hat{C}(1-u, 1-v) = 1-t \quad (39a)$$

$$(1-u)\hat{C}(1-u, 1-v) = 1-t, \quad (39b)$$

respectively, or, equivalently, the equations

$$(1-v)(1-u-v+C(u,v)) = 1-t \quad (40a)$$

$$(1-u)(1-u-v+C(u,v)) = 1-t, \quad (40b)$$

where $0 < t \leq 1$. Since equations (39a)–(39b) as well as equations (40a)–(40b) can be dealt with using the same techniques (u and v are exchangeable), below, we shall only investigate equations (39b) and (40b), i.e., those explaining the dynamics of V conditionally on the behavior of U . In general, finding the isolines of $\tau_{v|u}^{\geq}$ is a nontrivial task from an analytical point of view and may require the use of numerical routines. However, if either C or $\hat{C}$ are Archimedean, then the calculations may be simplified.

[72] Now, let $\hat{C}$ be Archimedean with generator γ ; we may then rewrite equation (39b) as

$$(1-u)\gamma^{[-1]}(\gamma(1-u) + \gamma(1-v)) = 1-t, \quad (41)$$

which yields the isoline $v = L_t(u)$, with

$$v = L_t(u) = 1 - \gamma^{(-1)}\left(\gamma\left(\frac{1-t}{1-u}\right) - \gamma(1-u)\right), \quad (42)$$

where $0 \leq u \leq 1 - \sqrt{1-t}$. Thus equation (42) defines the set in $\mathbf{I}^2$, where $\tau_{v|u}^{\geq}$ is constant and takes on the value

$$\tau_{L_t(u)|u}^{\geq} = \vartheta(t). \quad (43)$$

[73] Alternatively, suppose that C is Archimedean, with generator γ . Using again the variables a, b given by equation (34), we may then rewrite equation (40b) as

$$1-t = \left(1 - \gamma^{[-1]}(a)\right) \cdot \left(1 - \gamma^{[-1]}(a) - \gamma^{[-1]}(b) + \gamma^{[-1]}(a+b)\right), \quad (44)$$

i.e., a more tractable form, which only involves the monotone function $\gamma^{[-1]}$. Then, using equations (34) and (20), it is easy to find the isolines of constant return period in $\mathbf{R}^2$ of the conditional event $E_{y,x}^{Y|X}$ involving the pair (X, Y) as well as those of $E_{x,y}^{X|Y}$, simply by using the same approach and swapping u and v (or x and y).

4. Analyzing Some Case Studies

[74] In this section we shall illustrate an application of the techniques previously outlined by analyzing four case studies already examined in literature. In particular, we shall be concerned with hydrological variables of greatest interest in practice, i.e., flood volume, flood peak, storm (average) intensity, and storm duration. However, note that the same approach could be used to investigate, e.g., droughts.

4.1. Case of *Shiau* [2003]

[75] The first case study of interest is taken from *Shiau* [2003] (for a similar application, see also *Yue and Rasmussen* [2002]). The author uses the Type B bivariate extreme value distribution with Gumbel marginals to model extreme floods (characterized by flood peak and flood volume) in a basin in southern Taiwan. In this work the author does not use copulas; however, the cited bivariate distribution can simply be obtained by plugging suitable Gumbel marginals F_X, F_Y (i.e., considering $u = F_X(x)$ and $v = F_Y(y)$) into the expression provided by equation (8). In addition, equation (9) gives the formula of the corresponding generator. Incidentally, the same copula (with different marginals) and the same variables were also considered by *De Michele et al.* [2004] (see section 4.2).

[76] The author claims that the above bivariate distribution can be used to model the available data; however, no two-dimensional goodness-of-fit tests are shown to support such an assumption. As a matter of principle, other bivariate distributions (i.e., copulas other than the one cited) could provide valuable fits; note that such tests can easily be carried out using empirical copulas [*Nelsen*, 1999; *De Michele and Salvadori*, 2003; *De Michele et al.*, 2004; *Favre et al.*, 2004].

[77] In the cited paper the author plots the isolines of F_{XY} ; in Figure 4 we show its copula equivalent. Clearly, the copula approach is much more general than the one using a well-defined bivariate function F_{XY} . In fact, Figure 4 would be the same for all the distributions F_{XY} generated by the Gumbel-Hougaard copula, independently of the marginals adopted. Then, using equation (20), it is easy to find the

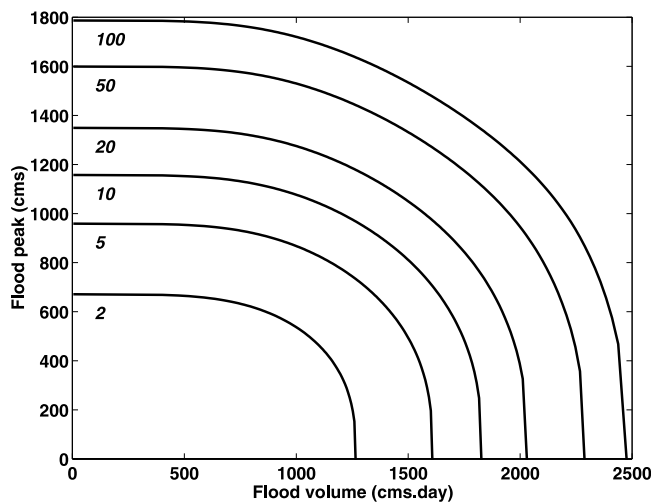


Figure 5. Level curves (isolines) of the conditional return period $\tau_{y|x}^{\vee}$ for the data considered in section 4.1. The values of the return periods are as indicated (in years), and the units are as by *Shiau* [2003].

corresponding isolines of F_{XY} in $\mathbf{R}^2$. In particular, since $\mathbf{C}$ is Archimedean, its level curves can be written explicitly via equation (14), i.e.,

$$v = L_{t_i}(u) = e^{-[(-\ln t_i)^\delta - (-\ln u)^\delta]^{1/\delta}}, \quad (45)$$

where $t_i \leq u \leq 1$ in Figure 4. Here $\delta = 1.294$; such a value was estimated exploiting a relation between δ and Pearson's linear correlation coefficient ρ . Although it is easy to calculate, we discourage the practitioners to use ρ as a means for estimating δ ; instead, we suggest using other quantities, such as Kendall's or Spearman's rank correlation coefficients. In fact, the latter ones always exist, whereas ρ depends upon the marginal laws F_X, F_Y and may not exist at all (e.g., when using suitable Pareto-like marginals (see *De Michele and Salvadori* [2003], *De Michele et al.* [2004], and also section 4.2)).

[78] In the cited paper the author also plots the conditional distributions

$$\mathbf{P}\{Y \leq y \mid X > x_\pi\} \quad \text{and} \quad \mathbf{P}\{X \leq x \mid Y > y_\pi\},$$

where X denotes the flood volume and Y the flood peak; here x_π and y_π are the marginal quantiles of X and Y , respectively, corresponding to the return period π . The author selects $\pi = 2, 5, 10, 20, 50, 100$ years. Such distributions can easily be obtained also by using copulas via equation (3). However, we believe that the knowledge of the conditional events $E_{y|x}^{Y|X}$ (or $E_{x|y}^{X|Y}$), having an assigned return period π , would be more justified and interesting for performing, e.g., flood risk assessment. Since μ_T is estimated as $\mu_T = 0.776$ years, using the results of section 3.4 and the six return periods mentioned above, it is easy to calculate the isolines of $\tau_{y|x}^{\vee}$. These are plotted in Figure 5.

[79] Finally, in the cited paper the author also plots the isolines of $\tau_{x,y}^{\vee}$ and $\tau_{x,y}^{\wedge}$ for the pair (X, Y) , using the same six thresholds mentioned above. In Figure 6 we show its copula equivalent. Again, the copula approach is much more general than the one used by the author since the level

curves in Figure 6 would be the same for all the distributions F_{XY} generated by the Gumbel-Hougaard copula, independently of the marginals adopted. Then, using equation (20), it is easy to find the corresponding curves in $\mathbf{R}^2$. In addition, since $\mathbf{C}$ is Archimedean, the isolines of $\tau_{u,v}^{\vee}$ can be derived from those of $\mathbf{C}$ and can be written explicitly using equation (45). As in the cited paper, the isolines of $\tau_{x,y}^{\wedge}$ for the pair (X, Y) would be bounded by the horizontal and the vertical axes, while those of $\tau_{x,y}^{\vee}$ would have no bounds (see, e.g., Figure 7 involving the same copula).

4.2. Case of *De Michele et al.* [2004]

[80] The second case study of interest is taken from *De Michele et al.* [2004]. The authors use a Gumbel-Hougaard copula (with upper-unbounded generalized extreme value marginals) to assess the spillway design flood of an existing Italian dam by considering the maximum annual flood volume and flood peak as the hydrological variables of interest (see also section 3.1).

[81] The authors estimate the dependence parameter $\delta = 3.055$ using Kendall's rank correlation coefficient. In fact, the existence of Pearson's linear correlation coefficient is

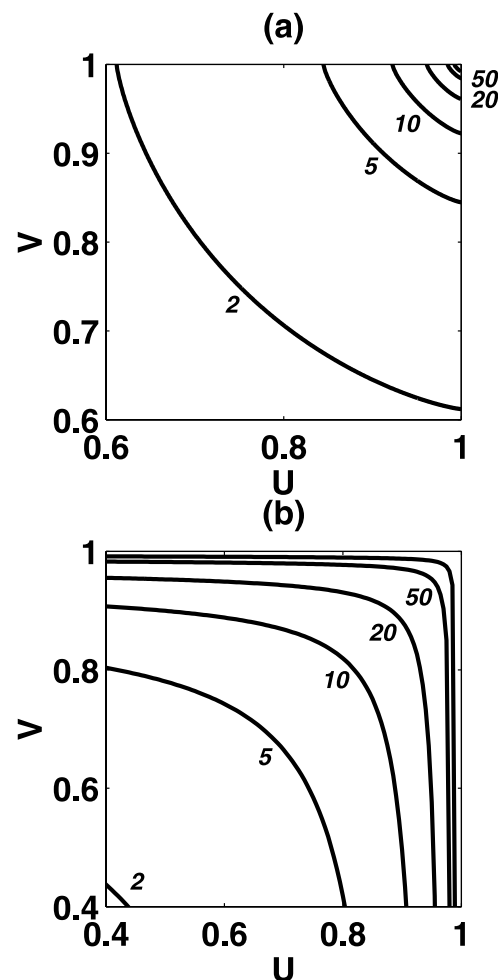


Figure 6. Level curves (isolines) of (a) $\tau_{u,v}^{\vee}$ and (b) $\tau_{u,v}^{\wedge}$ for the data considered in section 4.1. The values of the return periods (in years) are as indicated. In order to make the picture clear, the domains are subportions of $\mathbf{I}^2$, and the rightmost labels “100” are not drawn.

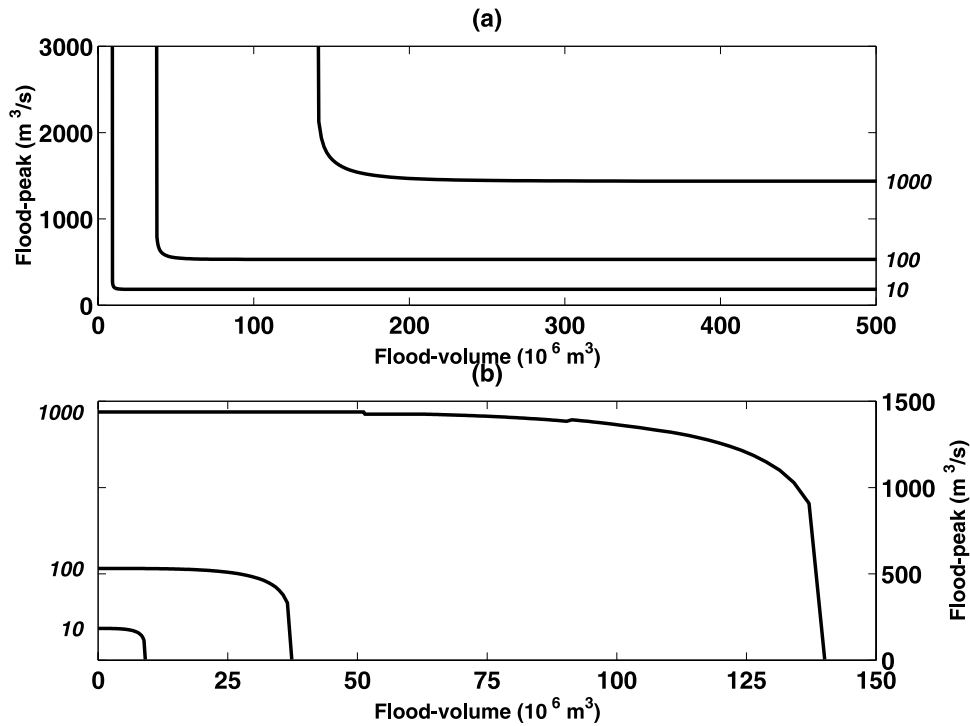


Figure 7. Level curves (isolines) of (a) $\tau_{x,y}^\vee$ and (b) $\tau_{x,y}^\wedge$ for the data considered in section 4.2. The values of the return periods (in years) are as indicated.

suspicious since the estimates of the marginal parameters indicate that the second-order moments may not exist.

[82] As an illustration, in Figure 7 we plot the isolines of $\tau_{x,y}^\vee$ and $\tau_{x,y}^\wedge$ for three selected thresholds (i.e., 10, 100, 1000 years). Here X denotes the maximum annual flood volume and Y the maximum annual flood peak. Clearly, the use of annual values yields $\mu_T = 1$ year.

[83] In section 3.1 we already discussed the differences between univariate and bivariate analysis; in particular, the results shown in Table 2 illustrate the practical limitations and drawbacks of the former approach. An important point is the calculation of the probability function K_C given by equation (27):

$$K_C(t) = t - \frac{1}{\delta} t \ln t, \quad (46)$$

where $t \in \mathbf{I}$. Note that the same analysis outlined below could be used to investigate the stochastic behavior of the phenomenon considered in section 4.1 since in both cases the copula belongs to the Gumbel-Hougaard family.

[84] Now, given an arbitrary critical level $t_* > 0$, it is immediate to calculate the secondary return period $q_{t_*}^\vee$ given by equation (30). As an illustration, in Figure 8 we plot the (survival) probability $\bar{K}_C(t)$ and the secondary return period $q_t^\vee$ as functions of $\vartheta(t)$; both the pictures are in a log-log plane in order to enhance the behavior of the functions plotted. Using $\vartheta(t)$ instead of t (obviously, with $t \in (0, 1)$) to parameterize the horizontal axis yields a direct evaluation of the probability that an event $E_{u,v}^\vee$, having a primary return period $\tau_{u,v}^\vee$ larger than $\vartheta(t)$, does show up at any given realization of the process as well as the value of the corresponding secondary return period (i.e., its average recurrence time).

[85] As a numerical illustration, in Table 3 we report some selected values of $\vartheta(t)$, $\bar{K}_C(t)$, and $q_t^\vee$. The last two columns provide the probability of a supercritical event at any given realization of the process (with respect to either the level t or the primary return period $\vartheta(t)$) and the expected time for such an event to show up, respectively. In particular, the first three lines give the probabilities and the expected times of supercritical events belonging to the regions to the right of the corresponding level curves shown in Figure 7a; without using copulas, such an important piece of information would be difficult to achieve.

4.3. Case of Favre et al. [2004]

[86] The third case study of interest is taken from Favre et al. [2004]. The authors use a copula taken from the Frank family (see equation (10) and the ensuing discussion) to model (1) the peak flow at a specific site, given by the combination of two different upstream flows, and (2) the joint behavior of peak flow and peak volume of a given river. In particular, the first case study shows how copulas may simplify the calculations when the marginals are finite mixtures of distributions, and the second one illustrates how different marginals (specifically, a Gumbel and a Gamma law) can be used to model the pair peak flow–peak volume. Indeed, Frank copulas may provide reasonable models for quite a few phenomena [see Hutchinson and Lai, 1990; Nelsen, 1999]; for some recent hydrological applications, see also De Michele and Salvadori [2003] and Salvadori and De Michele [2004].

[87] Apparently, the authors do not explore the whole potentialities of copulas for performing hydrological frequency analysis, a task which, instead, is fully developed and carried out in our work. Their conclusions are that univariate models may yield wrong indications for assessing

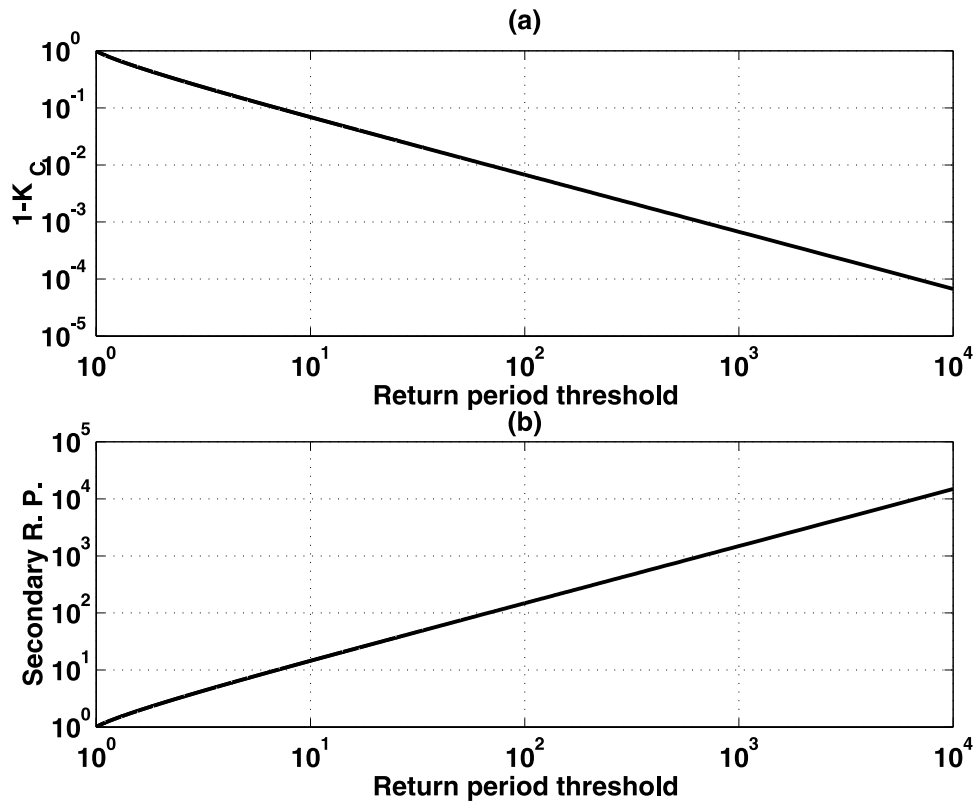


Figure 8. Plots of (a) the (survival) probability function $\bar{K}_C(t)$ and (b) the secondary return period function Q_t^V as functions of $\vartheta(t)$ for the data considered in section 4.2. The horizontal axes are in years as well as Q_t^V .

the hydrological risk, as we also pointed out in section 3.1. In particular, in their second case study the authors consider the conditional probability

$$p_{Y|X}(y) = \mathbf{P}\{Y \leq y \mid x_- \leq X \leq x_+\},$$

where Y is the peak volume and X is the peak flow, and the interval (x_-, x_+) corresponds to the 95% asymptotic confidence interval of the 100 year return period peak flow. Such a probability is of particular interest for water resources managers; as we already mentioned, the fundamental fact is that $p_{Y|X}$ can be calculated analytically by using equation (3).

[88] Given $0 < t \leq 1$, we may easily calculate the level curves of $\mathbf{C}$ (obviously, these are the same as of $\hat{\mathbf{C}}$ since $\mathbf{C} = \hat{\mathbf{C}}$):

$$L_t(u) = \frac{1}{\ln \delta} \ln \left(1 + (\delta - 1) \frac{\delta^t - 1}{\delta^u - 1} \right). \quad (47)$$

Thus, $\tau_{u,v}^\vee$, $\tau_{u,v}^\wedge$, and $\tau_{v|u}^\triangleright$ are constantly equal to $\vartheta(t)$ on

$$v = L_t(u), \quad (48a)$$

$$v = 1 - L_{1-t}(1 - u), \quad (48b)$$

$$v = 1 - L_{\frac{1-t}{1-u}}(1 - u), \quad (48c)$$

respectively. Note that these three sets of isolines may have different domains for the variable u (see equations (23), (32), and (42)). As an illustration, in Figure 9 we show the isolines of the Frank copula as well as those of the three return periods mentioned above for different values of the level t . In all the pictures, $\delta = 0.000112$, corresponding to the value of the dependence parameter estimated by the authors in their second case study. This means that the RVs involved are positively associated (as should be expected, dealing with peak flow and peak volume). Also note that for the variables considered, the authors use annual values; hence $\mu_T = 1$ year. The four pictures presented deserve some comments. First of all, the level t takes on the values indicated in each plot. In particular, the isoline corresponding to a specific value of t can be identified as follows: in Figures 9a and 9b it is the curve joining $(t, 1) - (1, t)$; in Figure 9c it is the curve joining $(0, t) - (t, 0)$; in Figure 9d it is the curve joining $(0, t) - (1 - \sqrt{1-t}, 0)$. Obviously,

Table 3. Selected Values of the (Survival) Probability Function $\bar{K}_C(t)$ and the Secondary Return Period Function Q_t^V for the Data Considered in Section 4.2

t	$\vartheta(t)$, years	$\bar{K}_C(t)$	Q_t^V , years
0.9	10	0.068961	14.5
0.99	100	0.006743	148.3
0.999	1000	0.000673	1486.3
0.9999	10,000	0.000067	14,865.8

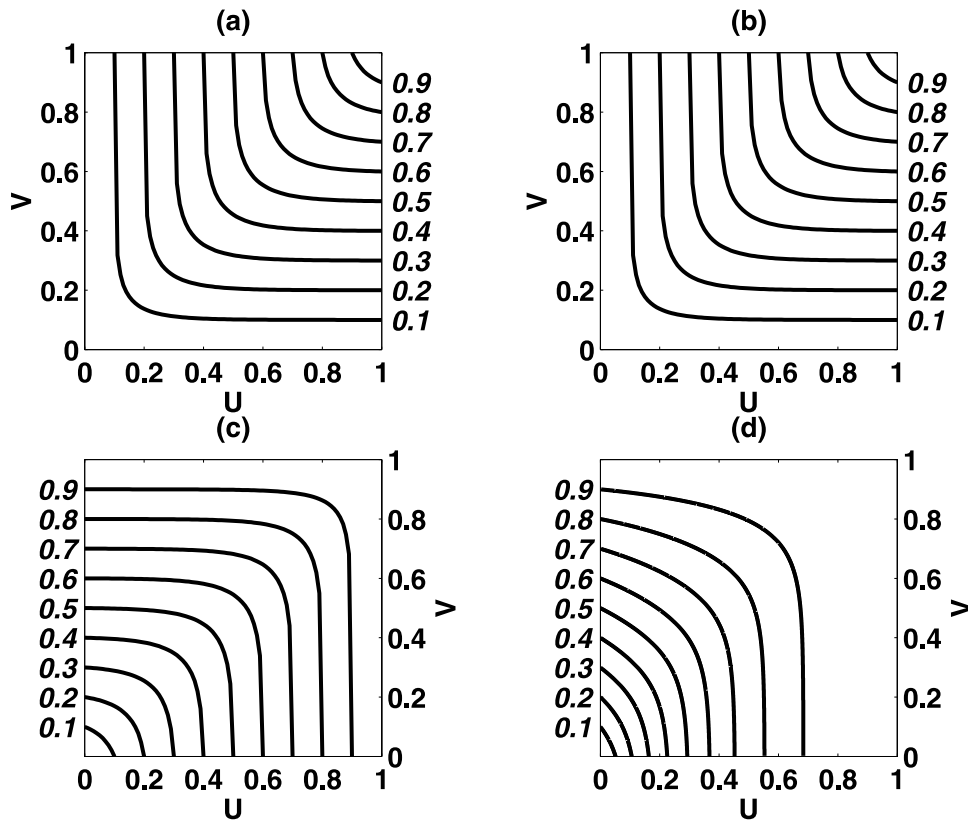


Figure 9. Level curves (isolines), for the data considered in section 4.3, of (a) the Frank copula, (b) $\tau_{u,v}^{\vee}$, (c) $\tau_{u,v}^{\wedge}$, and (d) $\tau_{u,v}^{>}$ (see text). The levels t are as indicated.

Figures 9a and 9b are identical. Furthermore, note the symmetries between Figures 9b and 9c: indeed, since $\mathbf{C} = \hat{\mathbf{C}}$, Figure 9b shows the (convex) level curves of $\mathbf{C}(u, v)$, while Figure 9c shows the (concave) level curves of $\hat{\mathbf{C}}(1-u, 1-v)$. The fact that the level curves have an almost L-shaped behavior has no particular meaning because it depends upon the copula considered and the scales of the x, y axes. Now, given arbitrary marginals F_X and F_Y (e.g., those considered by the authors), it is immediate to calculate the corresponding isolines, in the plane $\mathbf{R}^2$, for the return periods of the same events as expressed in terms of the RVs (X, Y) , e.g., the peak flow and the peak volume.

[89] An important point is the calculation of the probability function K_C :

$$K_C(t) = t - \frac{1}{\ln \delta} \delta^{-t} (\delta^t - 1) \ln \left(\frac{\delta^t - 1}{\delta - 1} \right), \quad (49)$$

where $t \in \mathbf{I}$. As an illustration, in Figure 10 we plot the (survival) probability $\bar{K}_C(t)$ and the secondary return period $Q_t^{\vee}$ as functions of $\vartheta(t)$. The same comments already expressed discussing Figure 8 hold also in the present case; in addition, Figure 10 should be compared to Figure 2, where the horizontal axis is parameterized simply by t .

4.4. Case of De Michele and Salvadori [2003]

[90] The last case study of interest is taken from *De Michele and Salvadori [2003]*. The authors use a Frank copula (with generalized Pareto marginals having negative shape parameters) to model the relation between the storm (average) intensity X and the storm duration Y in order to

calculate the probability distribution of the storm volume $P = XY$ under “extremal” conditions (i.e., when X is large and Y is small). This corresponds to considering extreme precipitations in short time periods, as is of interest in hydrologic design and natural hazards survey. (*Salvadori and De Michele [2004]*, again exploiting copulas, were also able to calculate analytically the unconditional distribution of P .)

[91] As an illustration, in Figure 11 we show the isolines of the joint distribution F_{XY} as well as those of the three return periods $\tau_{x,y}^{\vee}$, $\tau_{x,y}^{\wedge}$, and $\tau_{y|x}^{>}$ for five selected thresholds of the return periods. In all the pictures, $\delta = 2.271$, corresponding to the value estimated by the authors considering observed pairs with intensity $X > 1$ mm/h. This means that the RVs involved are negatively associated (as should be expected, dealing with storm intensity and duration); also, the authors estimated the average inter-arrival time between storms as $\mu_T = 146$ hours. The same comments already expressed discussing Figure 9 hold also in the present case; note that $X > 1$ mm/h in all four pictures and that the curves in Figures 11c and 11d are convex (this depends upon the derivatives of the marginals F_X and F_Y).

[92] Finally, the calculation of the functions $K_C(t)$ and $Q_t^{\vee}$ yields the pictures shown in Figure 12; the same comments already expressed discussing Figures 8 and 10 hold also in the present case. Note that the realization of events whose primary return period exceeds even small thresholds (say, $\vartheta(t) = 10$ years) is relatively quite rare, practically with a probability of the order of 10^{-6} , and the corresponding secondary return period is about 18,000 years. However, these results are only apparently surprising. In fact, in our

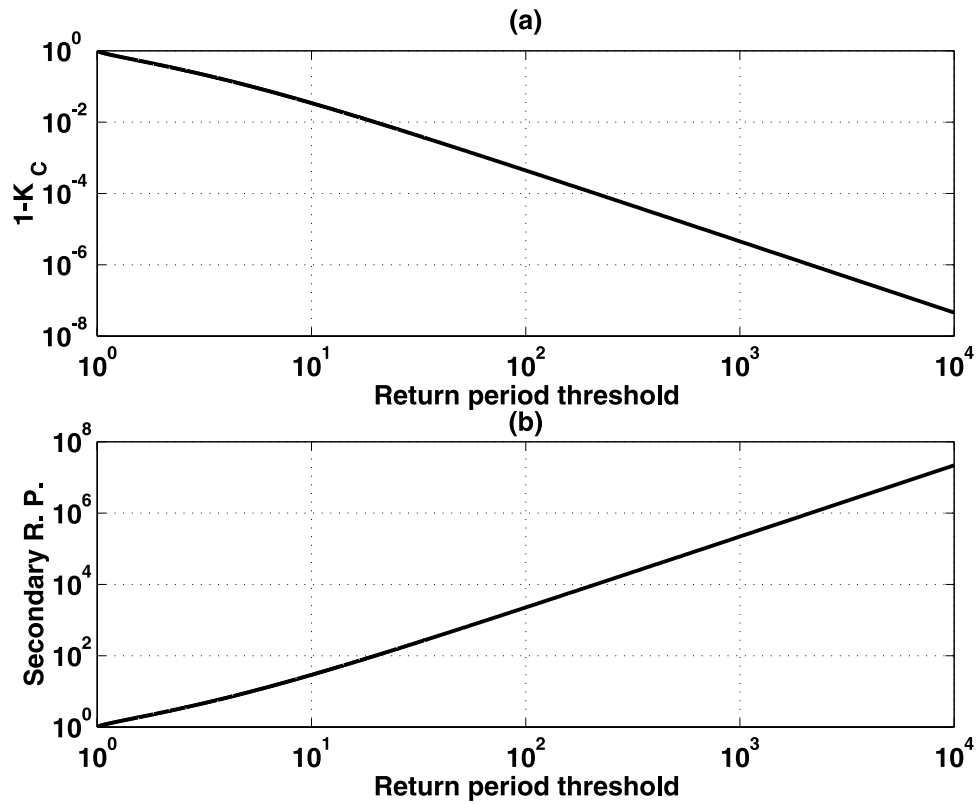


Figure 10. Plots of (a) the (survival) probability function $\bar{K}_C(t)$ and (b) the secondary return period function $\vartheta_t^\vee$ as functions of $\vartheta(t)$ for the data considered in section 4.3. The horizontal axes are in years as well as $\vartheta_t^\vee$.

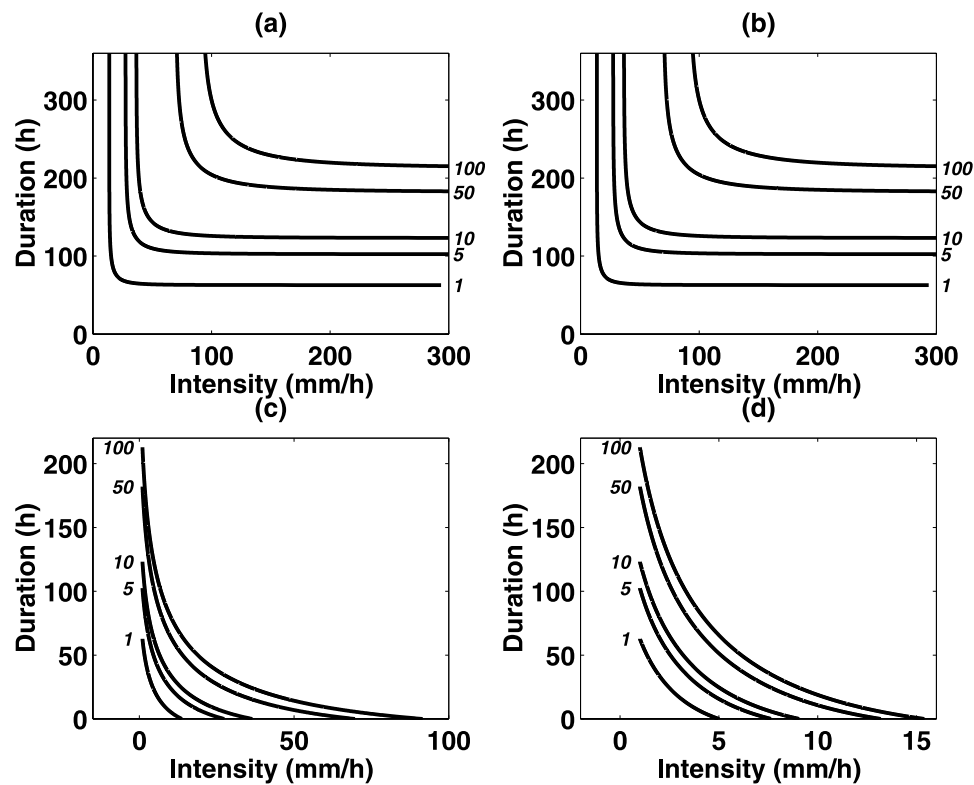


Figure 11. Level curves (isolines), for the data considered in section 4.4, of (a) the joint distribution $F_{X,Y}$ (b) $\tau_{x,y}^\vee$, (c) $\tau_{x,y}^\wedge$, and (d) $\tau_{y|x}^\wedge$ (see text). The values of the return periods are as indicated (in years).

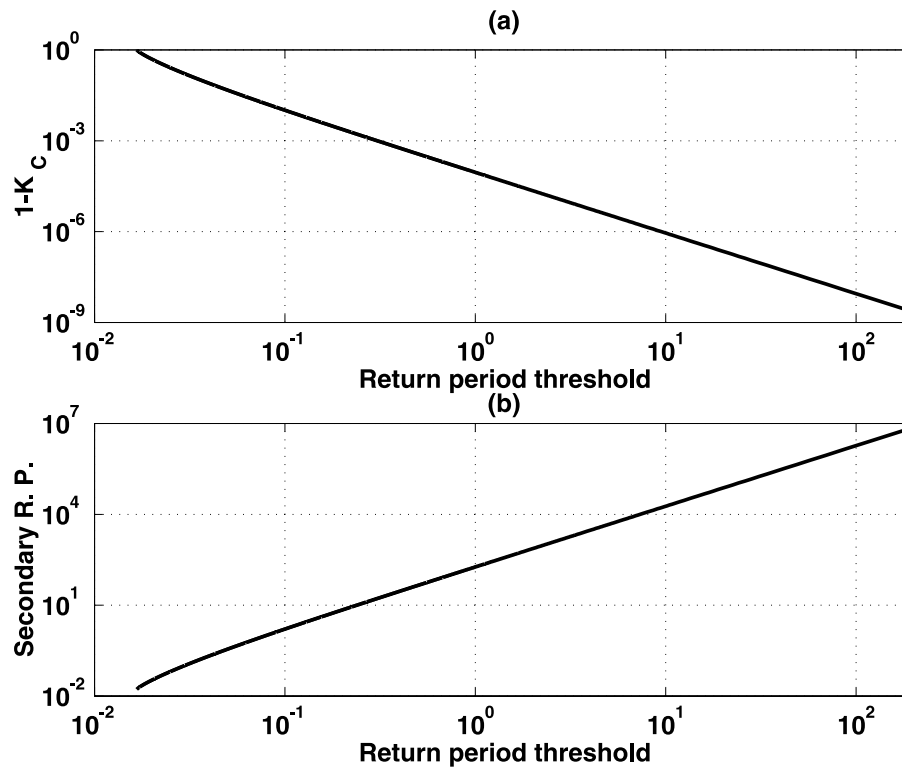


Figure 12. Plots of (a) the (survival) probability function $\bar{K}_C(t)$ and (b) the secondary return period function Q_t^V as functions of $\vartheta(t)$ for the data considered in section 4.4. The horizontal axes are in years as well as Q_t^V .

case the storms are quite frequent during the year ($\mu_T = 146$ hours, approximately one storm every 6 days); thus even looking for an event with a relatively small primary return period is equivalent to searching for a storm which happens on a temporal scale thousand of times larger than the fundamental (daily) scale of the rainfall process investigated. Indeed, a similar situation is illustrated in Figure 10, where the observations are on an annual basis, and more usual thresholds (say, $\vartheta(t) = 1000$ years) are associated with probabilities of the order of 10^{-5} . This scenario may be, at a first glance, nonintuitive, but the present approach is able to properly explain such unexpected results.

[93] As a numerical illustration, in Table 4 we report some selected values of $\vartheta(t)$, $\bar{K}_C(t)$, and Q_t^V ; the same comments already expressed discussing Table 3 hold also in the present case. In particular, the first three lines provide the probabilities and the expected times of supercritical events belonging to the regions to the right of the isolines labeled “1,” “10,” and “100,” respectively, in Figure 11b. Again, such an information is important during the design phase but would be difficult to obtain without using copulas.

5. Conclusions

[94] In this paper we outline a general theoretical framework exploiting copulas for studying the return period of hydrological events, which provides a very simple, yet efficient, means for doing risk analysis and represents a common criterion for sizing the manufactures. In particular, we consider events depending upon the joint behavior of two nonindependent random variables, an approach which

can easily be generalized to the multivariate case. As a relevant point, we make it clear that all the bivariate distributions present in literature can be described in a straightforward way in terms of suitable copulas.

[95] As a fundamental result, we show how the use of copulas may greatly simplify the calculations and even yield analytical expressions for the isolines of the return periods, both in the unconditional and in the conditional case. In particular, we show some of the most evident drawbacks and limitations of standard bivariate models not involving copulas, including the practical inadequacy to deal with complex marginals such as finite mixtures.

[96] Besides an exhaustive theoretical presentation and justification of the most relevant results, we thoroughly discuss the practical problems arising when performing hydrological risk assessment and show how the traditional univariate analysis may sometimes fail and yield an over/underestimation of the risk associated with given events.

[97] In addition, we show how a new, well-defined probability distribution may be associated with the return

Table 4. Selected Values of the (Survival) Probability Function $\bar{K}_C(t)$ and the Secondary Return Period Function Q_t^V for the Data Considered in Section 4.4

t	$\vartheta(t)$, years	$\bar{K}_C(t)$	Q_t^V , years
0.983333	1	0.0000907	183.6
0.998333	10	8.976×10^{-7}	18,569.1
0.999833	100	8.965×10^{-9}	1.859×10^6
0.999983	1000	8.538×10^{-11}	1.952×10^8

period of specific events and introduce the definitions of sub-, super-, and critical events as well as those of primary and secondary return periods. These functions may open new views and prospects in the field of frequency analysis since they provide easy formulas for dealing with the stochastic realization of potentially dangerous (destructive) events.

[98] An illustration of the techniques proposed is provided by analyzing four case studies already examined in literature; in particular, the hydrological variables involved are the ones of greatest interest in practice, i.e., flood volume, flood peak, storm (average) intensity, and storm duration. As a result, the investigation of the problems is simplified, and new information about the phenomena considered is gained, without requiring more effort than that spent by the authors cited.

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